

JAMES RUSE AGRICULTURAL HIGH SCHOOL

YEAR 12

1997 Mathematics 3u Term 2 Open Book Assessment

Time Allowed: 85 minutes

- Start each question on a new page
- Marks may not be awarded for poorly arranged work
- No equipment may be borrowed during the exam.

Question 1 (Start a new page)

(a) The rate of change in the number of birds (B) in a colony is given by $\frac{dB}{dt} = k(B - 1000)$ where t is in years and k is a constant. The number of birds at the start of 1990 was 1500.

(i) Show that $B = 500e^{kt} + 1000$

(ii) At the start of 1992 the number was 1800. Find the number at the start of 1997 to the nearest 100.

(b) A particle moves in a straight line so that at time t its displacement (x) from the origin is given by $x = t^3 - 2t^2 + 3t - 1$.

- By finding v (in terms of t) show that the particle never stops
- Find when it is travelling at its slowest speed (Justify your answer)

Question 2 (Start a new page)

(a) The letters of the word DEPOSITOR are to be arranged in a row. Find the number of arrangements if :-

- the "word" starts and ends with a consonant
- the vowels occupy the even places.

(b) The acceleration of a particle moving in a straight line is given by $\frac{d^2x}{dt^2} = 2x - 4$ where x is the displacement, in metres, from the origin O and t is the time in seconds. Initially the particle is at rest at $x = 4$

- If the velocity of the particle is v m/s, show that $v^2 = 2(x^2 - 4x)$
- Show that the particle does not pass through the origin.
- Find the position of the particle when $v = \sqrt{90}$. Justify your answer.

Question 3 (Start a new page)

A particle P is moving in SHM about a fixed point O. The displacement x metres at time t seconds is given by $x = 4\sqrt{2}\sin(3t + \frac{\pi}{4})$

- Find the equation for velocity in terms of x
- Show $\ddot{x} = -9x$
- Find the initial displacement and velocity of P
- Find the time taken to reach the maximum displacement from O, for the first time.

Question 4 (Start a new page)

(a) The letters A, B, C, D, E, F, G are arranged around a circle. Find the probability that A and B are not together.

(b) A daredevil on a motorbike is planning to jump a gorge 200m wide. At the gorge's edge the speed of the bike is 50m/s and the angle of projection is 30° above the horizontal.

- Write down without proof the cartesian equation of the trajectory.
- Show that the rider succeeds in jumping the river.
- Find the angle that the bike makes with the horizontal on the other side of the river.

Question 5 (Start a new page)

(a) The speed V (m/s) of a point moving along the x axis is given by

$$v^2 = 144x - 36x^2 + 180$$

- Prove the motion is SHM
- What is the maximum speed of the motion.

(b) A particle is projected with an initial speed of 64m/s (at an angle θ) towards a wall which is 100 m horizontally from the point of projection and 10 m high. (Air resistance is to be neglected and the acceleration due to gravity is taken as 10m/s^2 .)

- Show that the particle just clears the wall when $50000\tan^2\theta - 409600\tan\theta + 90960 = 0$
- In what range must the angle of projection lie for the wall to be cleared.

Question 6 (Start a new page)

(a) A total of 6 players is selected at random to form a volleyball team from a group of 10 girls and 5 boys. 6 of the girls and 3 of the boys have played the game before. Find the probability that of the 6 selected :-

- 4 are girls and 2 are boys.
- at least 4 of the players are girls that have played the game before.

(b) To lift a weight a boy uses a rope 20m long thrown over a tree branch which is 6m off the ground. He ties the rope around his waist (1m off the ground) and walks off at constant speed of 1m/s.

At time t, let the boy be distant x metres from the vertical below the weight and let the weight be y metres above the ground.

- Draw a diagram illustrating the above information
- Find how fast the weight is rising when it is 3m above the ground.

THIS IS THE END OF THE TEST

Q1

(a)

$$(i) B = 500e^{kt} + 1000$$

$$\frac{dB}{dt} = 500ke^{kt}$$

$$= k(B - 1000)$$

$$(iii) 1800 = 500e^{2k} + 1000$$

$$k = 0.235 \left(\frac{1}{2} \ln 8/5 \right)$$

$$\text{when } t = 7$$

$$0.235 \cdot 7$$

$$B = 500e^{1.665}$$

$$= 3590$$

$$= 3600$$

(Newton's law)

$$(b) x = t^3 - 2t^2 + 3t - 1$$

$$(i) V = 3t^2 - 4t + 3$$

which is pos def ($a > 0$)always > 0

$$(iii) a = 6t - 4$$

Speed up when $t = 2/3$

Up to that point acceleration is acting against motion. then slowing it + after $t = 2/3$ acceleration acting in same direction as motion thus acceleration is +

Q2 (a)

$$(i) \frac{5 \times 4 \times 7!}{2!} = 50400$$

$$(ii) \frac{5! \times 4!}{2} = 1440$$

$$(b) (i) \frac{d}{dt} \left(\frac{1}{2} v^2 \right) = 2x - 4$$

$$\frac{1}{2} v^2 = x^2 - 4x + C$$

$$C = 0$$

$$v^2 = 2(x^2 - 4x)$$

"Particle starts from rest at $x = 4$ & proceeds right ($a > 0$) until velocity is zero. it continues right. Now reaches 0.

$$(iii) \text{ when } v = 0$$

$$0 = 2(x^2 - 4x)$$

$$4x = x^2 - 4x$$

$$x^2 - 4x - 4x = 0$$

$$(x - 9)(x + 5) = 0$$

$$x = 9 \text{ or } -5$$

but it cannot be $x = -5$ for previous comment

$$x = 9$$

$$(i) \dot{x} = 12\sqrt{2} \cos(3t + \frac{\pi}{4})$$

$$= 12\sqrt{2} \sqrt{\cos^2(3t + \frac{\pi}{4})}$$

$$= \pm 12\sqrt{2} \sqrt{1 - \sin^2(3t + \frac{\pi}{4})}$$

$$= \pm 12\sqrt{2} \sqrt{1 - \frac{x^2}{32}}$$

$$= \pm 3\sqrt{32 - x^2}$$

$$(ii) v^2 = 9(32 - x^2)$$

$$\frac{1}{2} v^2 = \frac{9}{2} (32 - x^2)$$

$$\frac{d}{dt} \left(\frac{1}{2} v^2 \right) = -9x$$

$$(iii) t = 0 \quad x = 4$$

$$v = 12 - 12$$

Max displacement when $v = 0$

$$(iv) \text{ Displacement } = \sqrt{32} = 4\sqrt{2}$$

$$\sqrt{32} = 4\sqrt{2} \text{ m } (3t + \frac{\pi}{4})$$

$$1 = \sin(3t + \frac{\pi}{4})$$

$$t = \frac{\pi}{12} \text{ sec}$$

Q3

$$(i) \dot{x} = 12\sqrt{2} \cos(3t + \frac{\pi}{4})$$

Square both sides

$$v^2 = 144 \times 2 \cos^2(3t + \frac{\pi}{4})$$

$$v^2 = 288 (1 - \sin^2(3t + \frac{\pi}{4}))$$

$$v^2 = 288 (1 - \frac{x^2}{32})$$

$$v^2 = 288 - 9x^2$$

$$v = \pm 3\sqrt{32 - x^2}$$

$$(ii) x = 4\sqrt{2} \sin(3t + \frac{\pi}{4})$$

$$v = 12\sqrt{2} \cos(3t + \frac{\pi}{4})$$

$$\dot{x} = -12\sqrt{2} \sin(3t + \frac{\pi}{4}) \times 3$$

$$= -36\sqrt{2} \sin(3t + \frac{\pi}{4})$$

$$= -9[4\sqrt{2} \sin(3t + \frac{\pi}{4})]$$

$$= -9x$$

4

$$(a) \frac{4! \times 5 \times 4}{6!} = \frac{2}{3}$$

$$(b) (i) y = x \tan \alpha - \frac{g x^2}{2v^2} \left(\frac{1}{\cos^2 \alpha} \right)$$

$$(ii) 0 = \frac{x}{\sqrt{3}} - \frac{10x^2}{5000} \cdot \frac{4}{3}$$

$$= x \left(\frac{1}{\sqrt{3}} - \frac{4x}{1500} \right) = \frac{x}{\sqrt{3}}$$

$$x = \frac{1500\sqrt{3}}{12}$$

$$= 216.5 \text{ m}$$

range

$$(iii) y = \frac{\sqrt{3}x}{3} - \frac{4x^2}{1500}$$

$$y' = \frac{\sqrt{3}}{3} - \frac{8x}{1500} = 0$$

$$= 0.5773 \text{ where } x = 216.5$$

$$\tan \alpha = 0.5773$$

$$\alpha = 30^\circ \text{ (accept } 30^\circ \text{ as an answer)}$$

Q3

$$(iii) x = 4\sqrt{2} \sin(3t + \frac{\pi}{4})$$

When $t = 0$

$$x = 4\sqrt{2} \times \frac{1}{\sqrt{2}}$$

$$x = 4$$

$$v = 12\sqrt{2} \cos(3t + \frac{\pi}{4})$$

$$v = 12$$

(iv) Max displacement when $v = 0$

$$v = \sqrt{32 - x^2}$$

$$\therefore x = \sqrt{32}$$

$$\sqrt{32} = 4\sqrt{2} \sin(3t + \frac{\pi}{4})$$

$$1 = \sin(3t + \frac{\pi}{4})$$

$$\sin \frac{\pi}{2} = \sin(3t + \frac{\pi}{4})$$

$$3t = \frac{\pi}{4}$$

$$t = \frac{\pi}{12} \text{ sec}$$

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TERM 2

Q5

(a) (i) $V^2 = 144x - 36x^2 + 180$

$\frac{1}{2}V^2 = 72x - 18x^2 + 90$

$\frac{d}{dx} \left(\frac{1}{2}V^2 \right) = 72 - 36x$
 $= -36(x-2)$

which is of the form $-n^2(x-b)$
 $\therefore$ SHM

(ii) Max Speed occurs at centre of motion:

$V^2 = 144x - 36x^2 + 180$

$= 288 - 144x + 180 \quad \text{at } x=2$

$= 324$

$V = 18$

Max Speed = 18 m/s

(b) (i) $y = x \tan \alpha - \frac{gx^2}{2v^2} (1 + \tan^2 \alpha)$

$y=10 \quad x=100, \quad g=10$

$10 = 100 \tan \alpha - \frac{10 \cdot 10000}{2 \cdot (64)} (1 + \tan^2 \alpha)$

$10 = 100 \tan \alpha - \frac{50000}{4096} (1 + \tan^2 \alpha)$

$50000 \tan^2 \alpha - 409600 \tan \alpha + 90960 = 0$

(ii) $\tan \alpha = \frac{40960 \pm \sqrt{(40960)^2 - (4 \cdot 50000 \cdot 90960)}}{10000}$

$= \frac{40960 \pm 38675.59}{10000}$

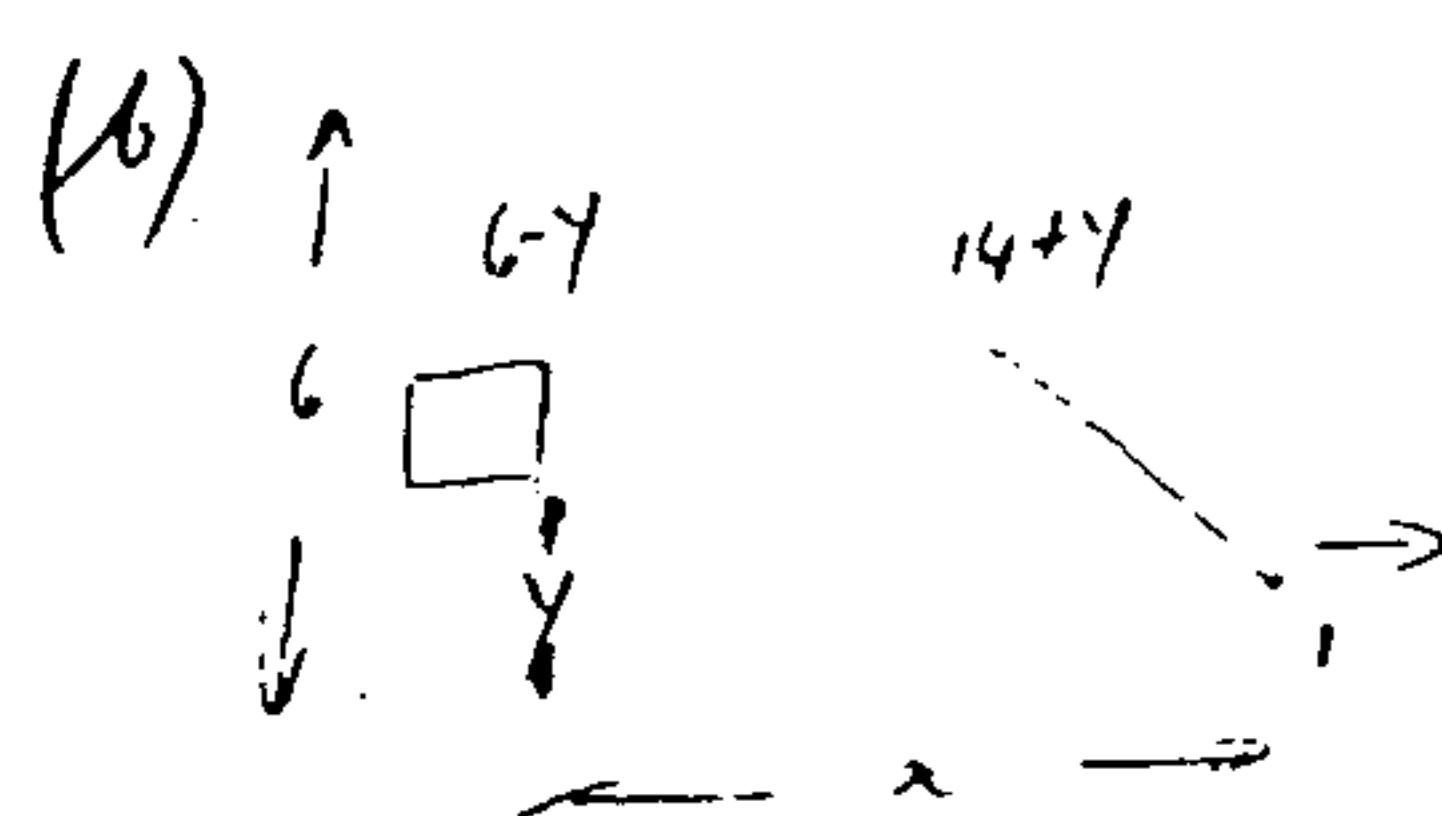
$\alpha = 82^\circ 50' \text{ or } 12^\circ 52'$

$\therefore$ the direction will $12^\circ 52' < \alpha < 82^\circ 50'$

Q6

(a) (i) $\frac{\binom{10}{4} \binom{5}{2}}{\binom{15}{6}} = \frac{60}{143}$

(ii) $\frac{\binom{6}{4} \binom{9}{2} + \binom{6}{5} \binom{9}{1} + \binom{6}{6}}{\binom{15}{6}} = \frac{17}{143}$



$(14+y)^2 = x^2 + 25$

$y = (x^2 + 25)^{\frac{1}{2}} - 14$

$y' = \frac{1}{2} (x^2 + 25)^{-\frac{1}{2}} \cdot 2x$
 $= \frac{x}{\sqrt{x^2 + 25}}$

when $y=3$

$(17)^2 = x^2 + 25$
 $x = \sqrt{264}$

$\frac{dy}{dx} = \frac{\sqrt{264}}{\sqrt{289}} = \frac{2\sqrt{66}}{17}$

$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$

$= \frac{2\sqrt{66}}{17} \times 1$

$= \frac{2\sqrt{66}}{17} \text{ m/s}$